

1. Introduction

- The ARM architecture has historically led mobile and embedded computing due to its power efficiency and strong ecosystem.
- However, RISC-V has emerged as an open-standard alternative, impacting geopolitics, supply chain sovereignty, and silicon economics.
- As Moore's Law slows, the industry increasingly requires domain specific architectures and deep hardware software co design.

2. Architectural Differences & Modularity

Both architectures rely on Reduced Instruction Set Computer (RISC) principles, but their fundamental approaches differ significantly.

ARM Architecture

- ARM uses an incremental, monolithic approach where licensees receive a fully defined, comprehensive feature set.
- This approach guarantees high standardization and software compatibility.
- However, it often forces designers to implement unneeded instructions and hardware blocks, resulting in silicon bloat.

RISC-V Architecture

- RISC-V is built on extreme modularity, starting with a tiny base integer instruction set.
- Additional functionalities are added as standard extensions, such as 'M' for multiply/divide or 'V' for vector operations.
- Hardware designers only implement necessary extensions, creating optimized, low-power, and small-footprint cores.

Vector Processing & Security

- **Vector Extensions:** ARM utilizes NEON and Scalable Vector Extension (SVE/SVE2), though SVE is restricted to high-end cores. RISC-V utilizes a variable-length vector model (RVV) that allows binary code to scale across different hardware register widths without needing recompilation.
- **Privilege & Security:** ARM relies on fixed exception levels and TrustZone for security. RISC-V offers a flexible privilege model and supports tailored, open-source trusted execution environments (TEEs) like Keystone and Penglai.

3. Licensing and Business Models

- **ARM (Proprietary):** ARM requires significant upfront licensing fees and ongoing royalties based on chip volume. This model creates high financial barriers and restricts custom modifications. While ARM recently introduced "Arm Custom Instructions" for limited modifications, it remains a tightly controlled framework.

- **RISC-V (Open-Standard):** Governed by RISC-V International, the ISA specifications are free and open-source. Designers can build and sell processors without paying royalties. This democratization lowers entry barriers and fosters collaborative tool development.

4. Ecosystem Maturity and Implementation

- **Software & Tools:** ARM benefits from decades of optimization, featuring auto-vectorization, security hardening, and dominance in operating systems like Android and iOS. RISC-V toolchains are advancing but lag in areas like mature auto-vectorization and polished OS distribution stacks.
- **Performance:** ARM offers pre-designed cores with predictable performance, power, and area (PPA) metrics. RISC-V ranges from open-source cores to proprietary IP, and while it struggles to match ARM's peak single-thread performance, it excels in power and area efficiency for custom accelerators.
- **Safety & Fragmentation:** ARM provides pre-certified safety packages for critical industries. RISC-V leaves safety validation up to the designer. Furthermore, RISC-V faces a fragmentation risk if proprietary extensions are added without being upstreamed, though this is mitigated by mandatory standard profiles.

5. Hardware-Software Co-design & DSAs

- **ARM Ecosystem:** Offers a stable, predictable environment with exceptionally optimized tools. However, deep customization is constrained because adding custom instructions usually requires a top-tier architectural license.
- **RISC-V Customization:** Enables profound hardware-software co-design by reserving a large opcode space for proprietary instructions. Designers can profile applications, define custom instructions, modify the RTL design, and co-simulate the hardware and software. This makes RISC-V inherently suited for building Domain-Specific Architectures (DSAs) as Moore's Law slows.

6. Strategic Impact and Future Outlook

- **Geopolitics:** RISC-V is seen as a "sovereign" architecture free from U.S. export embargoes, driving massive investments from Chinese tech giants and European initiatives.
- **Market Adoption:** As ARM explores subscription and value-based royalties, high-volume/low-margin markets (like IoT) are shifting toward RISC-V's royalty-free model.
- **Neutrality:** The blocked acquisition of ARM by NVIDIA heightened industry concerns about ARM's neutrality, boosting support for RISC-V as an independent alternative.